



NASA develops an atomic clock

The most precise timekeeping devices will now be used in deep space exploration. <https://digit.in/jul21-09>



Google's high-res map of brain

Google and Harvard published the largest high-resolution map of the of the human brain. <https://digit.in/jul21-10>

How do Neurotransmitters work?

Understand how the body sends messages within itself using the neurotransmitters as messengers.

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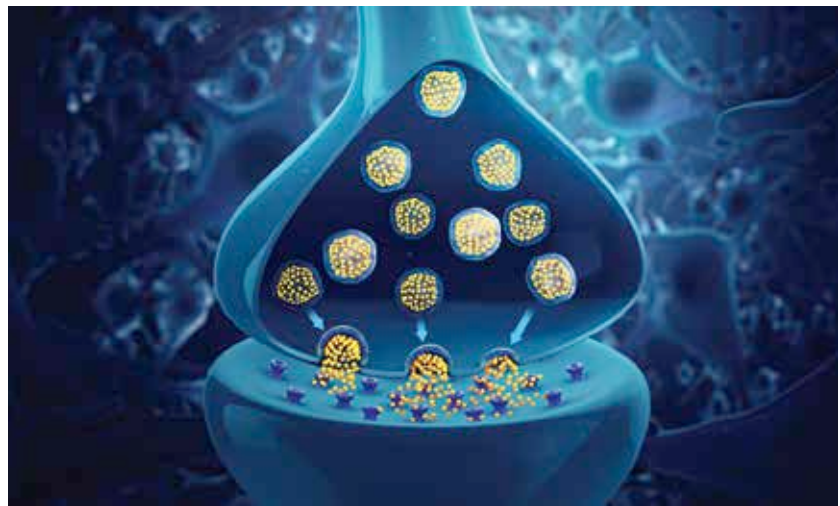
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here has been a long line of speculations regarding how the organs within the human body

communicate with each other. The answer is simple – neurotransmitters. Often called the body's chemical messengers, these chemicals help the body maintain a coordinated state of affairs between its organs, enabling us to perform coherent processes.

So, before telling you about the way neurotransmitters carry the messages within our body, it is important to provide a formal definition. As defined by the University of Queensland, Australia's researchers, neurotransmitters "are the molecules used by the nervous system to transmit messages between neurons, or from neurons to muscles." What they basically do is take the neural signals from one part of the body's complex system of neurons to the other.

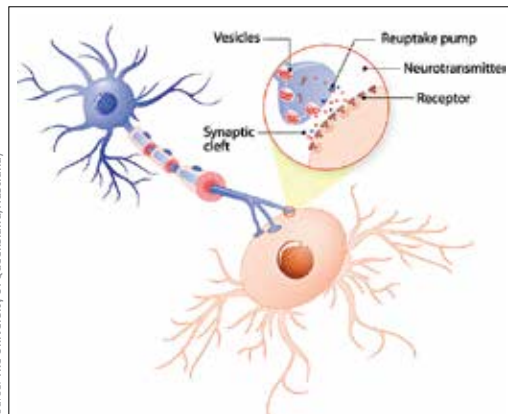
But there's a lot more to that process, and in this article we will have a look at that. So, you may know that all the signals transmitted in the human body are called nerve impulses. Everything that our brain tells our organs to do is through these



Synaptic vesicles containing neurotransmitters

electrical signals. Once this electrical impulse has been generated in the brain, it has to be carried through to other parts. And the channel through which they are transmitted is called a neuron. They are the specialised cells with a complex structure that are connected with each other and form a complex network in our body called the nervous system, which helps the

body execute all the things that go on within it and on the outside, like limb movements. You may ask now that what is the use of neurotransmitters in all of this when we have the neurons doing all the heavy lifting? There is. The work that neurotransmitters carry out in the whole process of an impulse being generated in the brain and then being transmitted



Source: The University of Queensland, Australia

Illustration of how neurotransmitters are transferred from one neuron to other

to the right organ is of establishing a clear connection between two neurons. There are a million neurons in the human body, and none of them is connected to each other physically. There is always a gap between the two ends of adjacent neurons, which is filled with neurotransmitters of various kinds.

Most neurotransmitters that are found in the human body are made up of one of the three things – small